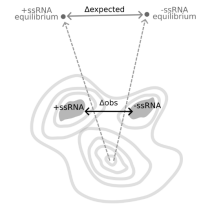
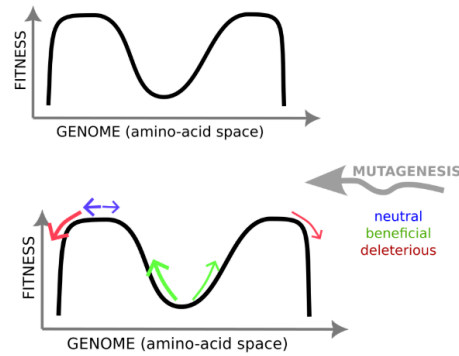
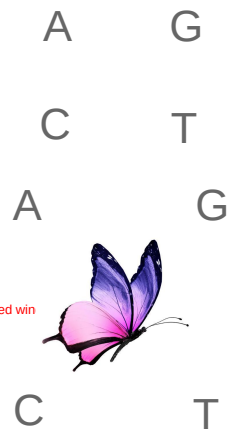


A biological “butterfly effect”: small shifts in mutational bias redirect long-term molecular evolution



biological state > mutagen exposure > mutational spectrum

genome composition > proteome composition

termite whale
longevity
non-termite cocoroashes mice
temperature/metabolism
coding single-stranded chain
UV exposure..

desamination of ss DNA adenine?

2 types of 12 comp MutSpec

syn codon usage
GAskew

	2 nd base			
	T ₁ A ₁	C ₁ C ₁	A ₁ A ₁	G ₁ G ₁
1 st base T ₁ T ₁	Phe TTT TTC	Tyr TCT TCA	Stop TAT TAA	Cys TGT TGC
C ₁ C ₁	Leu CTT CTC	Ser CCT CCA	His CAT CAA	Trp TGA TGG
G ₁ G ₁	Leu GTG GTC	Arg CGT CGA	Gln GAT GAA	Arg AGA AGG
A ₁ A ₁	Ile ATT ATA	Asn AAT AAA	Asp AAT AAG	Ser AGT AGC
G ₁ G ₁	Met ATA ATG	Lys AAT AAA	Asp AAT AAG	Ser AGT AGC
	Val GTA GTT	Gly GCT GCC	Glu GAT GAA	Gly GGT GGC

кузнечики?
mammals СУТВ?

gainer/loser in CYTB ~ generation length (egg size)